

Global Warming Science

<https://courses.seas.harvard.edu/climate/eli/Courses/EPS101/>

Acidification

Please use the template below to answer the workshop questions. “XX” indicates places where you need to complete/write code or add a discussion.

your name:

```
[ ]: # Plot carbonate species as C_T (DIC) varies:
import numpy as np
import matplotlib.pyplot as plt
import matplotlib
import pickle
import cartopy.crs as ccrs
import cartopy.feature as cfeature
from cartopy.feature import NaturalEarthFeature
from cartopy import config
from mpl_toolkits.axes_grid1 import make_axes_locatable

# load variables from a pickle file:
with open('./acidification_variables.pickle', 'rb') as file:
    d = pickle.load(file)
    # print information about each extracted variable:
    for key in list(d.keys()):
        print(f"extracting pickled variable: name={key}; size={d[key].shape}")
    globals().update(d)

# print ionization/ dissociation constants:
temp=15
sal=35
patm=1
depth=0
Kh=Kh.item()
K1=K1.item()
K2=K2.item()
Kw=Kw.item()
Kspc=Kspc.item()
Kspa=Kspa.item()
print(
    f"\nT={temp:g}, S={sal:g}, depth={depth:g}, "
    f"Kh={Kh:g}, K1={K1:g}, K2={K2:g}, "
    f"Kw={Kw:g}, Kspc={Kspc:g}, Kspa={Kspa:g}.\n"
)
```

Explanation of input variables

Time series of observed and projected CO₂ and time axes and the corresponding time axes:

CO₂_obs_global_decadal_mean, CO₂_obs_global_decadal_mean_years,

CO₂_rcp85_global_decadal_mean, CO₂_rcp85_global_decadal_mean_years

Time series of observed and projected global-mean ocean pH:

pH_obs_global_decadal_mean, pH_obs_global_decadal_mean_years,
pH_rcp85_global_decadal_mean, pH_rcp85_global_decadal_mean_years

pH maps: observed and projected:

pH_obs_1850_map, pH_obs_2010_map, pH_rcp85_2100_map,

the corresponding axes:

pH_map_latitude pH_map_longitude

equilibrium constants for the ocean carbonate system:

Kh: Henry's Law constant,

K1: first dissociation constant of H₂CO₃,

K2: second dissociation constant of H₂CO₃,

Kw: water dissociation constant

Kspa: solubility constant for CaCO₃ (aragonite),

Kspc: solubility constant for CaCO₃ (calcite),

Exact solution of the carbonate system as a function of pH for fixed Alkalinity: this is the solution obtained the full equations using a dedicated software (mocsy).

exact_alk, exact_co2, exact_co3, exact_hco3, exact_pH, exact_pco2

Comments:

- 1) Units are concentrations in millimol per kg (approximately millimol/liter).
- 2) exact_co2 is the dissolved CO₂, denoted H₂CO₃^{*} in the notes.
- 3) exact_pco2 is CO₂(g) in ppm.

1) Characterizing pH changes:

(a) Plot a time series of pH from 1850 to 2100, combining observations and RCP8.5 projection

```
[ ]: # plot:
plt.figure(figsize=(12,5))
#plt.plot(XX)
#plt.plot(XX)
plt.xlabel("year")
plt.ylabel("pH")
plt.legend();
```

(b) Calculate the percent change in [H⁺] concentration involved in the pH change observed so far from 8.16 to 8.06.

```
[ ]: XX
```

c) Plot pH maps for 1850, 2010 (observed) and 2100 (rcp8.5):

```
[ ]: # plot contours for 1850:
plt.figure(dpi=200)
levels=np.arange(7.2,8.4,0.01)
c=plt.contourf(pH_map_longitude, pH_map_latitude, pH_obs_1850_map,levels)
# draw the colorbar
clb=plt.colorbar(c, shrink=0.85, pad=0.02)
# add title/ labels:
```

```

plt.xlabel('Longitude')
plt.ylabel('Latitude')
clb.set_label('pH')
plt.title('pH observed 1850')
plt.show()

# plot contours for 2010:
#XX
plt.title('pH observed 2010')
plt.show()

# plot contours:
# XX
plt.title('pH rcp8.5 2100')
plt.show()

```

d) Discuss the amplitude of current spatial variations versus expected temporal changes in the context of the expected robustness of ocean biology to acidification. XX

e) Calculate and contour the percentage change in $[H^+]$ concentration due to pH changes from year 1850 to 2010 and to 2100.

```

[ ]: #H_1850 = XX # pH is -Log10[H+]
      #H_2010 = XX # pH is -Log10[H+]
      # Contour the percentage change
      #XX

plt.title('Percentage change in [H+] between 1850 and 2010')
plt.show();

```

2) The carbonate system solution

a) Plot the given precise solution of the carbonate species CO_3^{2-} , HCO_3^- , H_2CO_3 and the total CO_2 , as function of pH, for the case of fixed Alk and varying C_T .

```

[ ]: # print sample values in order to identify range of valid approximation:
print("\n\n      pH,      TOT_CO2,      pco2")
totCO2_exact=exact_hco3+exact_co3+exact_co2
for i in range(0,len(exact_pH),10):
    # add more variables to print as needed:
    print(f"\t{exact_pH[i]:.4f} \t\t{XX}")

# plot the exact solution for the different species using different colors:
plt.plot(exact_pH,exact_hco3,label="HCO$_3^-$")
plt.plot(XX,XX,label="CO$_3^{2-}$")
plt.plot(XX,XX,label="H$_2$CO$_3$") # this is H2CO3*
plt.plot(XX,XX,label="$\Sigma$CO$_2$")
plt.plot(XX,XX,label="Alk")

plt.xlim(XX)
plt.ylim(XX)
plt.xlabel("pH")
plt.legend();

```

2b) Plot separately the pH and carbonate ion as function of atmospheric CO_2 from the same solution.

```
[ ]: # set defaulty font size:
fig=plt.figure(figsize=(5,3),dpi=300)
# plot pH:
axis1=plt.gca()
axis2 =axis1.twinx()
h1=axis1.plot(XX,XX,color="b",label="pH")
axis1.set_ylabel("pH",color="b")
axis1.set_xlabel("CO$_2$(g) (ppm)")
axis1.set_ylim(XX,XX)
axis1.set_xlim(XX,XX)

# plot CO$_3^{2-}$:
h2=axis2.plot(XX,XX,color="r",label="CO$_3^{2-}$")
axis2.set_ylabel("CO$_3^{2-}$ (milimole/liter)",color="r")

h = h1+h2
legend_labels = [l.get_label() for l in h]
axis1.legend(h, legend_labels, loc="upper right")

plt.tight_layout()

plt.show()
```

3) The carbonate system buffer effect:

plot the expected change to the total dissolved CO₂ (that is, to C_T) as a function of the atmospheric CO₂ concentration in the range 200 to 1000 ppm, based on the given exact solution. Superimpose the expected change calculated based on Henry's law alone (in this case, to H₂CO₃* only), that would have occurred without the carbonate system reactions. Discuss the role of the buffer effect in mitigating the anthropogenic increase atmospheric CO₂.

```
[ ]: # understanding buffer capacity
print(f"Henri's constant: Kh={Kh:g}; Kh*280/1000={Kh * 280 / 1000:g}")
fig=plt.figure(figsize=(7,3),dpi=300)

plt.subplot(1,2,1)
# -----
# plot dissolved CO2 vs atmospheric pCO2 in both scenarios:
# -----
plt.plot(exact_pco2,XX,color="b",label="CO$_2$(aq)")
# need to convert pCO2 to ppt for calculation given units of Kh and CO2(aq):
CO2_aq_withouot_buffer=XX
plt.plot(exact_pco2,CO2_aq_withouot_buffer,color="g",label="CO$_2$(aq) w/o buffer")
plt.ylabel("CO$_2$(aq) mm/l")
plt.xlabel("CO$_2$(g) (ppm)")
plt.ylim(0,0.05)
plt.xlim(100,1000)
plt.grid(linewidth=0.25)
plt.legend()

plt.subplot(1,2,2)
# -----
```

```

# plot changes to total dissolved CO2 in both scenarios relative to state at pCO2=280:
# -----
# find total CO2 from the plot in the answer to question 2:
totCO2_exact_at_280=XX
plt.plot(exact_pco2,XX,color="b",label="$\\Delta\\Sigma$CO$_2$")
# need to convert pCO2 to ppt for calculation given units of Kh and CO2(aq):
Delta_CO2_aq_withouot_buffer=(XX
plt.plot(exact_pco2,Delta_CO2_aq_withouot_buffer,color="g",label="$\\Delta$CO$_2$(aq) w/
  ↳o buffer")
plt.ylabel("dissolved/total CO$_2$ (mm/l)")
plt.xlabel("CO$_2$(g) (ppm)")
plt.ylim(-0.3,0.3)
plt.xlim(100,1000)
plt.grid(linewidth=0.25)
plt.legend()

plt.tight_layout()

plt.show()

```

Discussion:

XX.

4) Understanding the response using the approximate solution to the carbonate system:

- (a) Explain what are the conditions for the approximate solution discussed in section 5.2.2 to be valid, and identify the range of atmospheric CO₂ concentrations for which you expect these conditions to hold.

XX

- (b) Use the approximate solution of the carbonate system to calculate the concentration of the carbonate species as a function of pH for Alk=2.2 mM/l, in the range of atmospheric CO₂(g) in which you expect the approximate solution to be valid. To do so, vary the C_T over an appropriate range of values, and calculate both the carbonate species and pH. Superimpose dashed lines representing the provided exact solution.
- (c) Plot the approximate solution for the pH as function of atmospheric CO₂(g). In both cases, superimpose a plot of the exact solution.

```

[ ]: def solve_approximate_carbonate_system(C_T,alk):
    """Calculate the approximate solution of carbonate system, solving
    for pH,pco2(=CO2(g)), co2 (=dissolved CO2=H2CO3*), hco3 and co3 as
    a function of total CO2 and alkalinity.

    Inputs are in milimole per liter,
    outputs same, output pco2 is in ppt, needs to be multiplied
    by 103 to get it in ppm.

    The carbonate system reaction constants K1,K2,Kh are loaded with the
    data at the beginning of the workshop and are therefore known to
    this function.
    """
    hco3=XX

```

```

co3=XX
H=XX
pH=-np.log10(H)
co2=XX # this is  $H_2CO_3^*$ 
pco2=XX # this is  $CO_2(g)$ 

return pH,pco2,co2,hco3,co3

# reference values for inputs for approximate solution:
C_T0=2000e-3
alk0=2300e-3
C_T_range=C_T0*np.arange(0.9,1.15,0.01)
print(
    f"\nconsidering C_T over the range of : "
    f"{C_T_range[0] * 1000} {C_T_range[-1] * 1000} micro mole/liter"
)
N=len(C_T_range)
# initialize array of output values:
approx_alk=np.zeros(N)
approx_pH=np.zeros(N)
approx_pco2=np.zeros(N)
approx_co2=np.zeros(N)
approx_hco3=np.zeros(N)
approx_co3=np.zeros(N)
i=-1
for C_T in C_T_range:
    i=i+1
    pH,pco2,co2,hco3,co3 = solve_approximate_carbonate_system(alk=alk0, C_T=C_T)
    approx_alk[i]=alk0
    approx_pH[i]=pH
    approx_pco2[i]=pco2
    approx_co2[i]=co2 # this is  $H_2CO_3^*$ 
    approx_hco3[i]=hco3
    approx_co3[i]=co3
    #print(pH,pco2,fco2,co2,hco3,co3)

# plot calculated approximate solution as function of pH:
plt.figure(1,figsize=(8,5),dpi=200)
approx_totCO2=approx_hco3+approx_co3+approx_co2
plt.plot(approx_pH,approx_hco3,label="HCO $_3^-$ ",lw=3)
#plt.plot(XX,XX,label="CO $_3^{2-}$ ",lw=3)
#plt.plot(XX,XX,label="H $_2$ CO $_3^*$ ",lw=3) # this is  $H_2CO_3^*$ 
#plt.plot(XX,XX,label="$\Sigma$CO $_2$ ",lw=3)
plt.plot([8.2,8.2],[-0.1,3.0],color="k",linewidth=0.5,linestyle="--",label="pre-indust_
    _pH")
#plt.plot(XX,XX,"--",label="alk",lw=3)
plt.xlabel("pH")
plt.legend()

# plot exact solution as dotted yellow line:
totCO2_exact=exact_hco3+exact_co3+exact_co2
plt.plot(exact_pH,exact_hco3,":y",label="exact")

```

```

plt.plot(exact_pH, exact_co3, ":y")
plt.plot(exact_pH, exact_co2, ":y") # this is H2CO3*
plt.plot(exact_pH, totCO2_exact, ":y")
plt.plot(exact_pH, exact_alk, ":y")
plt.xlim(6.5, 9.5)
plt.ylim(-0.1, 3.0)
plt.legend()
plt.tight_layout()

# plot pH vs co2(g):
plt.figure(2, figsize=(8, 5))
#XX note units of approx_pco2, and convert to ppm for plot:
#plt.plot(XX, XX, "-r", label='approximate pco2, ppm')
#plt.plot(XX, XX, ":y", label='exact pco2')
plt.ylabel('pH')
plt.xlabel('CO$_2$(g)')
plt.title("approx and exact pH as function of atmospheric CO$_2$(g)")
plt.xlim(100, 1000)

plt.legend()
plt.grid(lw=0.25)
plt.ylim(7.6, 8.6)
plt.tight_layout()

```

- (d) A heuristic understanding of the carbonate ion decline for increased atmospheric CO₂(g): First, use the approximate solution of section 5.2.2 to calculate the concentration of all carbonate system ions at a CO₂(g) of 280 ppm and for a 10% increase (308 ppm). (Hint: assume $Alk = 2.2$ mM/l, find the C_T value that results in each of the two CO₂(g) concentrations, and solve using this value). Next, given an increase of 10% in atmospheric CO₂(g), use Henry's law to show that carbonic acid, [H₂CO₃*] increases by 10% as well. Then use the equation for the first dissociation to calculate the response of the bicarbonate and H⁺ ions. (Hint: the increased concentration of carbonic acid leads to X more mM/l of bicarbonate and X of H⁺; use the equilibrium equation between 1.1 [H₂CO₃*] and its products to derive a quadratic equation for X and solve it.) Finally, taking the calculated values of H⁺ and HCO₃⁻, use the equation for the second dissociation to calculate the response of the carbonate ion. Explain your results for [H₂CO₃*], [HCO₃⁻], and [CO₃²⁻], compare to the full solution of the approximate system for CO₂(g) of 308 ppm, and discuss the limitations of this heuristic approach.

Solution: XX

Limitations: XX

```

[ ]: # solving for concentrations before increase in atmospheric CO2:
alk=2.2
# to calc C_T:
# (1) use the C_T curve in the plot for question 4c;
# (2) as a check, sum the relevant carbon species taking values from the same above_
    -> plot
C_T=XX
pH, pco2, co2, hco3, co3 = solve_approximate_carbonate_system(C_T, alk)
H=10**(-pH)
print(
    f"before perturbation (full approximate solution):\n"
    f" alk={alk:.6g}, C_T={C_T:.6g}, pco2={pco2 * 1.e3:.6g}, "
    f"H={H:.6g} ppm, pH={pH:.4f}\n"

```

```

f" co2,hco3,co3={co2 * 1.e3:.6g}, "
f"{hco3 * 1.e3:.6g}, {co3 * 1.e3:.6g}"
)

# after a CO2(g) increase, solving for X in first dissociation:
# equation is  $X^2 + bX + c = 0$ , solution is  $X = (-b \pm \sqrt{b^2 - 4c})/2$ 
b=XX; c=XX
X1=(-b+np.sqrt(b**2-4*c))/2 # we expect a positive X, so negative root is ignored
X2=(-b-np.sqrt(b**2-4*c))/2 # we expect a positive X, so negative root is ignored
print(f"X1,X2={X1}, {X2}")
X=1.0*X1
H_after_1st=H+X
pH_after_1st=-np.log10(H_after_1st)
hco3_after_1st=hco3+X # bicarbonate
print(
    f"after 1st dissociation: "
    f"H_after_1st,pH_after_1st={H_after_1st:g}, {pH_after_1st:g}"
    f"\n hco3_after_1st={hco3_after_1st * 1.e3:g}"
)

# solving for second dissociation:
Y = XX
print(f"Y={Y:g}")

co3_after_2nd = co3 + Y # carbonate
print(
    f"after 2nd dissociation:"
    f"\n co3_after_2nd={co3_after_2nd * 1.e3:g}"
)

# compare to approximate solution with increased CO2(g) of 308 ppm:
alk=2.2
C_T=XX
pH,pco2,co2,hco3,co3 = solve_approximate_carbonate_system(C_T,alk)
H=10**(-pH)
print(
    f"after perturbation (full approximate solution):\n"
    f" alk={alk:g} C_T={C_T:g}, pco2={pco2 * 1.e3:g}, "
    f"H={H:g} ppm, pH={pH:g}"
    f"\n co2,hco3,co3={co2 * 1.e3:g} {hco3 * 1.e3:g} {co3 * 1.e3:g}"
)

```

5) Estimating observed and future pH from CO₂:

5a) Use the approximate solution with $Alk = 2.2$ mM/liter, varying C_T to calculate the pH and CO₂(g). Find the C_T values that result in the observed CO₂ concentrations at years 1850 and 2000 and the RCP8.5 value at 2100. Deduce the pH values at these years. Compare to the above ocean pH time series for 1850–2100.

```

[ ]: print("year, pCO2, year, pH")
      print("-----")

for i in range(len(CO2_obs_global_decadal_mean)):
    print(

```



```

        f"{CO2_obs_global_decadal_mean_years[i]:g} "
        f"{CO2_obs_global_decadal_mean[i]:g} "
        f"{pH_obs_global_decadal_mean_years[i]:g} "
        f"{pH_obs_global_decadal_mean[i]:g}"
    )

print("-----")

for i in range(len(CO2_rcp85_global_decadal_mean)):
    print(
        f"{CO2_rcp85_global_decadal_mean_years[i]:g} "
        f"{CO2_rcp85_global_decadal_mean[i]:g} "
        f"{pH_rcp85_global_decadal_mean_years[i]:g} "
        f"{pH_rcp85_global_decadal_mean[i]:g}"
    )

```

```

[ ]: # now calculate CO2:
alk = 2.2

year = 1850.0 # 285 ppm
C_T = 2.0
pH, pco2, co2, hco3, co3 = solve_approximate_carbonate_system(C_T, alk)
print(
    f"year={year:g} alk={alk:g} C_T={C_T:g}, "
    f"pco2={pco2 * 1.e3:g} ppm, pH={pH:g}"
)

year=2000.0 # 369 ppm
C_T=XX
XX
XX

year=2100.0 # 935 ppm
C_T=XX
XX
XX

```

5b) Optional extra credit: Use a similar procedure to the approximate solution of the carbonate system equation in section 5.2.2 to solve for the pH given the Alk and CO₂(g) (instead of using given Alk and C_T as in that section). The solution involves deriving a quadratic equation for [H⁺] whose coefficients include Alk and CO₂(g) and the various reaction coefficients. Let Alk=2.2 mM/l and plot the expected upper ocean pH given the observed CO₂ concentration values from 1850 to present and the RCP8.5 values from present to 2100. Compare to the given pH values for these periods obtained from measurements and more sophisticated modeling by superimposing a plot of the given surface ocean pH for 1850–2100. Solution: Start with the

approximate carbonate equations,

$$\begin{aligned}
 K_H &= \frac{[\text{H}_2\text{CO}_3^*]}{[\text{CO}_2(\text{g})]}, \\
 K_1 &= \frac{[\text{HCO}_3^-][\text{H}^+]}{[\text{H}_2\text{CO}_3^*]}, \\
 K_2 &= \frac{[\text{CO}_3^{2-}][\text{H}^+]}{[\text{HCO}_3^-]}, \\
 \text{Alk}_C &= [\text{HCO}_3^-] + 2[\text{CO}_3^{2-}], \\
 C_T &= [\text{HCO}_3^-] + [\text{CO}_3^{2-}].
 \end{aligned} \tag{1}$$

In this case we are given $\text{CO}_2(\text{g})$ and Alk_C . starting to solve we have from the first equation,

$$[\text{H}_2\text{CO}_3^*] = \text{XX}$$

and then from the second,

$$[\text{HCO}_3^-] = \text{XX}$$

we can now write the third equation as

$$K_2 = \frac{[\text{CO}_3^{2-}][\text{H}^+]}{K_1 K_H [\text{CO}_2(\text{g})]/[\text{H}^+]}$$

so that,

$$[\text{CO}_3^{2-}] = \text{XX}$$

The (specified) alkalinity can now be written as

$$\text{Alk}_C = \text{XX}$$

which we can get a quadratic equation for this concentration,

$$\text{XX}[\text{H}^+]^2 - \text{XX}[\text{H}^+] - \text{XX} = 0$$

Because the CO_2 and the alkalinity are known, all coefficients are known and we can solve for $[\text{H}^+]$ and thus for the pH.

```
[ ]: def solve_approximate_pH_from_Alk_and_CO2g(alk,pCO2):
    """
    Calculate the approximate solution of the carbonate system, solving for the pH
    as function of pco2(=CO2(g)) and alkalinity.
    Input Alk is in millimole per liter, input pCO2 is in ppt (=ppm/1000).
    """
    # The quadratic equation for pH is ax^2+bx+c=0, where x=[H^+]:
    XX=np.nan # remove this line after replacing XX with proper variables and
    ↪ expressions
    a=XX
    b=XX
    c=XX
    H1=(-b+np.sqrt(b**2-4*a*c))/(2*a)
    # H2=(-b-np.sqrt(b**2-4*a*c))/(2*a) # this is the non-physical solution
    pH=-np.log10(H1)
    co2=XX # this is H2CO3*
    hco3=XX
    co3=XX
    C_T=XX
```

```

    return pH,co2,hco3,co3,C_T

# calculate expected pH from observed and RCP8.5 pCO2:
# -----
pH_from_CO2_obs_global_decadal_mean=CO2_obs_global_decadal_mean*0.0
for i in range(0,len(pH_from_CO2_obs_global_decadal_mean)):
    pH,co2,hco3,co3,C_T = solve_approximate_pH_from_Alk_and_CO2g \
        (alk,CO2_obs_global_decadal_mean[i]/1.0e3)
    pH_from_CO2_obs_global_decadal_mean[i]=pH

pH_from_CO2_rcp85_global_decadal_mean=CO2_rcp85_global_decadal_mean*0.0
for i in range(0,len(pH_from_CO2_rcp85_global_decadal_mean)):
    pH,co2,hco3,co3,C_T = solve_approximate_pH_from_Alk_and_CO2g \
        (alk,CO2_rcp85_global_decadal_mean[i]/1.0e3)
    pH_from_CO2_rcp85_global_decadal_mean[i]=pH

# plot:
# -----
plt.figure(figsize=(12,5),dpi=200)
plt.plot(pH_obs_global_decadal_mean_years,pH_from_CO2_obs_global_decadal_mean
        ,"x",color="blue",label="pH from CO$_2$ obs")
plt.plot(pH_rcp85_global_decadal_mean_years,pH_from_CO2_rcp85_global_decadal_mean
        ,"+",color="red",label="pH from CO$_2$ RCP8.5")
plt.plot(pH_obs_global_decadal_mean_years,pH_obs_global_decadal_mean
        ,"o",color="blue",markerfacecolor='none',label="pH obs")
plt.plot(pH_rcp85_global_decadal_mean_years,pH_rcp85_global_decadal_mean
        ,"o",color="red",markerfacecolor='none',label="pH RCP8.5")
plt.xlabel("year")
plt.ylabel("pH")
plt.title("pH from obs/rcp8.5 compared to approximate solution from CO$_2$")
plt.legend();

```

```
[ ]:
```